Preamble

EART60071 Measuring and Predicting 2

Numerical modelling is important in many aspects of environmental science. Models are used to improve our understanding of natural systems, to interpret observations, and to inform decisions and policy. Examples range from assessing pollution in an area to investigating how the atmosphere and climate respond to changing conditions.

Many of the processes encountered in this module are described by *dynamical models*, based on systems of coupled differential equations. Some simple equations can be solved analytically, but realistic coupled systems often cannot. We therefore use approximate, but accurate, numerical methods to solve them on a computer.

Observations are critical to science: they are used to test theories and to develop new ideas. In the real environment, however, it is often impossible to control all variables in the way that we could in a laboratory experiment. A purely observational study may also sample only a limited set of conditions. Numerical models provide a complementary approach because they allow controlled experiments in which one aspect of a system can be changed while other factors are held fixed.

Some real-world experiments would also be impractical, expensive or ethically difficult. One example is *marine cloud brightening*, the proposal to seed low marine clouds with sea-salt aerosol so that they reflect more solar radiation (Latham, 1990). Before considering an intervention of this kind, we would want calculations that explore whether the intended effect is plausible and what unintended effects might occur. Models and simulations provide one way of carrying out those controlled tests.

In the first practicals you will learn the basic workflow used throughout the module: connect to the course Linux server, obtain model code, edit model inputs, run a calculation, and inspect or transfer the resulting output. It can look unfamiliar initially, but the same pattern recurs throughout the course.

What will I learn?

The module develops transferable computational and analytical skills alongside more specific environmental-science applications. You will practise:

- basic Linux commands and remote computing;
- downloading and updating model code using Git and GitHub;
- using Python to plot and inspect model data;
- interpreting complex model output, including output that varies in space and time;
- identifying relationships between variables and connecting those relationships to physical processes.

Specific examples in the module include:

- finite-difference methods for solving equations;
- cloud formation on aerosol particles;
- transport of pollutants in the atmosphere;
- processes governing the formation of rain from clouds;
- tsunamis and large-scale atmospheric or oceanic dynamics.

What this course is not

This is not intended to be a course in constructing a complete numerical model from scratch. Building models requires substantial experience in programming and numerical algorithms. We will discuss these ideas and look at model equations and code where they help our understanding, but the main emphasis is on using, interpreting and evaluating environmental models. Students who want more detail on numerical algorithms may find Hoffman (1992) useful.

References

Hoffman, J. D. (1992). *Numerical Methods for Engineers and Scientists.* McGraw-Hill.

Latham, J. (1990). Control of global warming? *Nature*, 347, 339–340. https://doi.org/10.1038/347339b0